

StockCare

Predictive Demand & Priority-Aware Distribution
Network for Tunisia's Pharmacies

A closed-loop AI pipeline linking pharmacy shortage forecasting,
national medicine prioritization, and optimized last-mile delivery



Phase 1 — The Idea

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Team Chaneb+

Adam Sabri HLAOUA — IMT Atlantique

Mohamed Mounib HALOUANI — Faculté de Pharmacie de Monastir

Omar AMARA — IMT Atlantique

Mohamed Aziz NCIR — ENIT

Malek Taieb NABLI — Ponts et Chaussées

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1 Executive Summary

Tunisia's pharmaceutical supply chain is reactive by design: pharmacies restock based on habit and gut feeling, depots allocate scarce medicine on a first-come-first-served basis, and delivery trucks follow routes drawn from experience rather than data. The result, documented extensively in the national press over the last three years, is chronic, recurring stockouts of essential medicines even when stock exists somewhere in the system [1,2,3].

StockCare closes this loop with three connected AI/OR components:

1. A **pharmacy-side forecasting engine** that predicts short-term medicine demand from contextual signals (region, season, calendar, exam periods, heatwaves) and compares it to real stock to auto-generate restock requests.
2. A **depot-side priority engine** that scores every medicine in the national catalogue by clinical and logistical criticality, so scarce supply and delivery capacity are allocated to what matters most first.
3. A **MILP-based routing engine** that sequences deliveries to minimize distance, fuel, and CO₂, while explicitly rewarding faster delivery of high-priority, cold-chain, or life-critical orders.

Nothing like this closed loop – forecasting → prioritization → priority-aware routing – currently exists in Tunisia's distribution chain. This document lays out the problem, the system design, the underlying models and math, the projected impact, and how the team is splitting the work.

2 The Problem

2.1 A chronic, structural shortage crisis

Medicine shortages in Tunisia are not an occasional inconvenience – they are a standing feature of the health system. Reporting places the number of unavailable products at times above 500 references across hospitals and private pharmacies [1], and pharmacists' union representatives have repeatedly cited a rolling shortage of 250 to 300 medicines at any given moment [2]. As recently as January 2026, pharmacists were still flagging stock that had not been replenished for over two months and escalating the issue to the national medicines agency [3]. A 2026 sector analysis (BMI/Fitch Solutions) warns that the combination of Pharmacie Centrale de Tunisie (PCT) debt, dinar depreciation, and foreign-currency restrictions on pharmaceutical imports is structurally pushing medicines toward being simultaneously *scarcer and more expensive* [4,5].

2.2 Why the system keeps failing

The financial and macroeconomic causes (PCT debt, CNAM reimbursement delays, forex shortages) are real, but they are compounded – and made far worse in their day-to-day effect on patients – by an **information and logistics problem** that is entirely solvable with data:

- **No demand anticipation.** Pharmacies order reactively, once shelves are already empty. Predictable demand spikes – flu season, summer heatwaves, university exam periods driving demand for specific OTC and chronic-treatment categories – are not modeled at all.
- **No visibility for the depot.** When dozens of pharmacies request the same scarce medicine simultaneously, the depot has no systematic way to decide who should receive the limited stock first based on clinical urgency.
- **No optimized delivery.** Once an order is approved, the trip from depot to pharmacy is planned

manually. A life-critical, cold-chain-sensitive order can sit behind lower-urgency stops purely because of geographic convenience, wasting the time, fuel, and CO₂ budget that scarce deliveries cannot afford to lose.

Key insight

The shortage crisis has a financial root cause that a student project cannot fix. But a large share of its *patient-facing damage* – the wrong medicine being out of stock at the wrong pharmacy at the wrong time – is a forecasting, prioritization, and routing problem. That is exactly what StockCare targets.

3 Proposed Solution — System Overview

StockCare is a three-layer closed loop connecting every pharmacy's stock and point-of-sale system to the depot's procurement and delivery fleet.

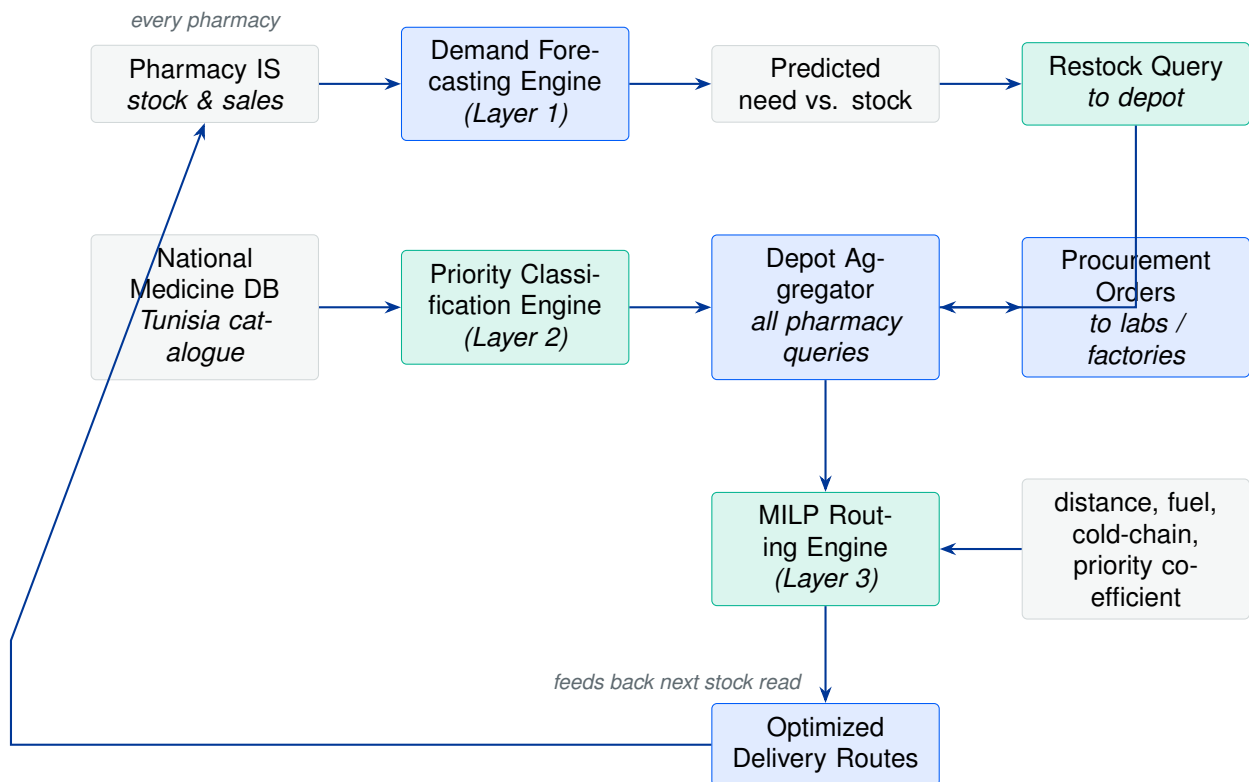


Figure 1: StockCare end-to-end pipeline: pharmacy forecasting → depot prioritization → priority-aware routing, closing the loop back to pharmacy stock.

3.1 Layer 1 — Pharmacy-Side Demand Forecasting Engine

For every (pharmacy, medicine) pair, the model forecasts expected demand over a rolling horizon H (e.g. the next two weeks) and compares it to live stock pulled from the pharmacy's information system.

Modeling approach: a gradient-boosted tree ensemble (e.g. LightGBM/XGBoost) over engineered lag and calendar features, or a hybrid statistical + ML model (seasonal decomposition combined with a residual learner), is a good fit given tabular, mixed categorical/numerical, medium-frequency data. The output is a per-medicine demand forecast $D_{i,m}$ for pharmacy i ,

Feature group	Examples	Why it matters
Geography	Region / governorate	Disease prevalence, climate, and demographics vary sharply across Tunisia
Calendar	Season, month, day of week, hour	Captures weekly and seasonal consumption cycles (respiratory illness in winter, etc.)
Academic context	Is-exam-period flag	Spikes in demand for stimulants, anxiolytics, painkillers, energy/vitamin products
Climate context	Heatwave / weather-alert flag	Spikes in rehydration salts, antihistamines, sunburn and GI treatments
Epidemiological signal	Flu-season index, outbreak alerts	Anticipates surges tied to seasonal epidemics
History	Lagged sales (7/14/30 days), stock-outs history	Core autoregressive signal for any demand model
Live state	Current stock level, pending orders	Needed to compute the actual shortage gap, not just raw demand

Table 1: Feature set for the pharmacy-level demand forecasting model.

medicine m , over horizon H .

Shortage-gap logic:

$$G_{i,m} = \max(0, D_{i,m} + S_{i,m} - X_{i,m}) \quad (1)$$

where $S_{i,m}$ is a safety-stock buffer and $X_{i,m}$ is the current stock read from the pharmacy's information system. Whenever $G_{i,m}$ exceeds a configurable threshold τ , the engine automatically emits a restock query

$$Q_{i,m} = \{ \text{pharmacy_id} = i, \text{medicine_id} = m, \text{quantity} = G_{i,m}, \text{urgency}, \text{timestamp} \}$$

sent to the depot in real time – no manual phone calls or paper order forms.

3.2 Layer 2 — Depot-Side Priority Classification Engine

The depot receives restock queries from many pharmacies simultaneously. Before anything can be routed, the system needs to know *which medicine, for which pharmacy, matters most right now* – this is the role of the priority engine, which draws on the national AMM registry maintained by the Direction de la Pharmacie et du Médicament (DPM), the Ministry of Health body registering every medicine legally sold in Tunisia: at extraction time it held **6,058** active AMMs (4,280 génériques, 1,695 princeps, 67 biosimilaires) [8] – this is the scale of catalogue Layer 2 needs to ingest and classify.

Conceptually, each medicine request is scored as:

$$P_{i,m} = \alpha \cdot \text{Criticality}(m) + \beta \cdot \text{StockoutRisk}(i, m) + \gamma \cdot \text{Irreplaceability}(m) + \delta \cdot \text{ColdChainSensitivity}(m) + \varepsilon \cdot \text{PopulationImpact}(i, m) \quad (2)$$

with weights normalized ($\alpha + \beta + \gamma + \delta + \varepsilon = 1$) and $P_{i,m} \in [0, 1]$, where:

- **Criticality** – is the medicine used for a chronic or life-threatening condition (e.g. insulin, cardiovascular, oncology) vs. comfort/OTC use;

- **StockoutRisk** – how close pharmacy i already is to running out, and how long it has been short;
- **Irreplaceability** – whether an equivalent generic/substitute is available nearby;
- **ColdChainSensitivity** – whether the medicine requires refrigerated transport, making delay itself a risk of spoilage;
- **PopulationImpact** – how many patients at that pharmacy depend on it (e.g. chronic-treatment renewals).

This score feeds two downstream decisions: **(a)** how the depot orders from labs and factories (higher-priority medicines are procured first and in larger safety margins), and **(b)** the routing engine described next.

The engine is split into two connected sub-components:

Static scoring – a learned reference table from the AMM registry. The registry’s fields map almost directly onto the static components of the score: substitute counts per DCI + dosage yield *Irreplaceability* (a princeps with no registered generic scores high), the galenic form (*Forme / Présentation*) flags likely cold-chain requirements, and the DCI is the join key for an ATC therapeutic-class enrichment separating chronic/vital from comfort use. A sample of the enriched registry is annotated by pharmacists with reference criticality scores, and a gradient-boosted regressor generalizes these annotations to the full catalogue. Because the source is public, official, and regulator-maintained, the resulting scoring table is reproducible and auditable by a third party – a benchmark other solutions would need to match, rather than a black-box heuristic. *Stock-outRisk* and *PopulationImpact* come from each pharmacy’s transactional history (Layer 1), not from the registry.

Adaptive weighting – reinforcement-learning-based regulation. The weights $(\alpha, \beta, \gamma, \delta, \varepsilon)$ should not stay fixed: a seasonal epidemic should push up Criticality and ColdChainSensitivity for vaccines; an import disruption at one laboratory should push up Irreplaceability for its DCIs. An RL agent regulates the weights over time from aggregated context (stockout rates by category, cold-chain alerts, seasonal indicators, import-delay signals), rewarded on post-decision KPIs (chronic-patient stockouts avoided, cold-chain waste avoided, critical-order delay) and penalized for induced overstocking. Because exposing an untrained agent to live shortage decisions is unacceptable in healthcare, it is trained offline first on a digital twin of the pharmacy network, then deployed with bounded step sizes, human validation above a threshold, and automatic fall-back to the static weights if the reward signal becomes anomalous. Crucially, only the weights become time-varying – the score’s form and the $pr_i \in [0, 1]$ interface exposed to Layer 3 stay identical, keeping the swap transparent to the MILP.

Current status

Both sub-components are scheduled for the next development iteration. Until they are delivered, Layer 3 (the MILP) treats the priority coefficient as a fixed constant (e.g. a 3-tier static weight: chronic/vital, essential, comfort) so that routing development can proceed in parallel and simply swap in the dynamic score once it is ready – the interface (a per-order priority value $pr_i \in [0, 1]$) is designed to make that swap trivial.

3.3 Layer 3 — MILP-Based Delivery Routing Engine

Once orders are prioritized, the depot must decide the order in which its heterogeneous fleet visits pharmacies. This is a variant of the **Vehicle Routing Problem (VRP)**, formulated as a

Mixed-Integer Linear Program (MILP) so that priority, distance, fuel, SLA deadlines, and cold-chain constraints are optimized jointly rather than handled by ad-hoc rules. The formulation below is the consolidated Phase-1 model.

Sets

- $N = \{1, \dots, n\}$: pharmacies to be served; node 0: the depot.
- $K = \{1, \dots, k\}$: the heterogeneous fleet of delivery vehicles.

Temporal & geographic parameters

- $d_{ij} \in \mathbb{R}^+$: distance between nodes i, j (km); $t_{ij} \in \mathbb{R}^+$: direct travel time between i, j (hours).
- $\sigma_i \in \mathbb{R}^+$: unloading time required at pharmacy i (hours).
- $H_{max} \in \mathbb{R}^+$: end of the planning horizon (e.g. 12h).
- $T_i \in \mathbb{R}^+$: target SLA delivery deadline for pharmacy i .

Vehicle & order parameters

- $Q_k \in \mathbb{R}^+$: maximum capacity of vehicle k ; $R_k \in \{0, 1\}$: 1 if vehicle k is refrigerated.
- $c_f^k \in \mathbb{R}^+$: usage cost (fuel/emissions) per km of vehicle k – now vehicle-specific to reflect a heterogeneous fleet.
- $q_i \in \mathbb{R}^+$: total load required for pharmacy i 's order.
- $cc_i \in \{0, 1\}$: 1 if order i requires the cold chain.
- $pr_i \in [0, 1]$: normalized priority score of order i – **fixed constant for now**, later supplied dynamically by Layer 2.

Objective weights & constants

- $w_p \in \mathbb{R}^+$: penalty weight for SLA lateness (dinars/hour).
- $\gamma \in \mathbb{R}^+$: a very small tie-break coefficient for latency.
- $w_v \in \mathbb{R}^+$: flat penalty for mobilizing an additional vehicle.
- $M_{ij} \in \mathbb{R}^+$: an adjusted Big- M constant, defined as $M_{ij} = H_{max} + \sigma_i + t_{ij}$ – tighter than a generic constant, which helps the solver.

Decision variables

- $x_{ijk} \in \{0, 1\}$: 1 if vehicle k travels directly from node i to node j .
- $z_k \in \{0, 1\}$: 1 if vehicle k is mobilized (leaves the depot).
- $s_i \in \mathbb{R}^+$: exact arrival time at node i .
- $L_i \in \mathbb{R}^+$: cumulative lateness at node i relative to T_i .

Objective

$$\min \underbrace{\sum_{k \in K} \sum_{i, j} c_f^k d_{ij} x_{ijk}}_{\text{fuel / CO}_2} + \underbrace{\sum_{i \in N} w_p \cdot pr_i \cdot L_i}_{\text{SLA lateness}} + \underbrace{\gamma \sum_{i \in N} pr_i \cdot s_i}_{\text{tie-break}} + \underbrace{w_v \sum_{k \in K} z_k}_{\text{fleet size}} \quad (3)$$

- **Fuel / CO₂** – cost of fuel and emissions across every leg actually driven.
- **SLA lateness** – the time overrun beyond each order’s target deadline, weighted by that order’s priority: being late on a high-priority order costs far more than being late on a low-priority one.
- **Tie-break** – a deliberately tiny coefficient γ on arrival time itself, so the objective doesn’t go “flat” once all lateness terms hit zero; it nudges the solver to still deliver urgent orders as early as possible, not just on time.
- **Fleet size** – a flat cost per vehicle mobilized, keeping the number of trucks and drivers used to a minimum.

Every term is expressed in dinars, so the weights w_p, γ, w_v stay interpretable and negotiable with the depot’s operations team rather than being abstract tuning knobs.

Constraints

$$\sum_{k \in K} \sum_j x_{ijk} = 1 \quad \forall i \in N \quad (\text{single service}) \quad (4)$$

$$\sum_j x_{jik} = \sum_j x_{ijk} \quad \forall i, \forall k \quad (\text{flow conservation}) \quad (5)$$

$$\sum_{j \in N} x_{0jk} = z_k \quad \forall k \quad (\text{fleet activation}) \quad (6)$$

$$\sum_{i \in N} q_i \left(\sum_j x_{ijk} \right) \leq Q_k \cdot z_k \quad \forall k \quad (\text{capacity}) \quad (7)$$

$$\sum_j x_{jik} \leq R_k \quad \forall i : cc_i = 1, \forall k \quad (\text{cold chain}) \quad (8)$$

$$s_j \geq s_i + \sigma_i + t_{ij} - M_{ij}(1 - x_{ijk}) \quad \forall i, j, k \quad (\text{time \& subtours}) \quad (9)$$

$$L_i \geq s_i - T_i \quad \forall i \quad (\text{SLA overrun}) \quad (10)$$

$$L_i \geq 0 \quad \forall i \quad (\text{non-negativity}) \quad (11)$$

$$s_0 = 0 \quad (\text{depot clock}) \quad (12)$$

- **Single service** – every pharmacy is visited exactly once.
- **Flow conservation** – any vehicle entering a node must also leave it.
- **Fleet activation** – z_k switches on as soon as vehicle k leaves the depot, linking it to the fleet-size penalty above.
- **Capacity** – a vehicle’s total load can never exceed Q_k , and tying the bound to z_k forces $z_k = 1$ the moment it carries any load.
- **Cold chain** – a pharmacy whose order needs the cold chain can only be reached by a refrigerated vehicle.
- **Time & subtours** – propagates arrival time leg by leg (including unloading time σ_i) using a tightened Big- $M_{ij} = H_{max} + \sigma_i + t_{ij}$; because time strictly increases along any used arc, this single family of constraints also rules out disconnected sub-loops that never pass through the depot – no separate MTZ variables needed.
- **SLA overrun & non-negativity** – lateness is the (possibly zero) overshoot past the target deadline T_i ; arriving early never produces negative “credit”.

- **Depot clock** – the schedule starts at $s_0 = 0$, anchoring every downstream arrival time.

In plain terms: the solver still wants short, fuel-efficient routes across a fleet of vehicles with different costs, but every hour a high-priority order runs past its SLA is penalized far more than the same lateness on a low-priority order – and even when nothing is technically late, the tie-break term still nudges urgent deliveries to happen first. That is exactly the trade-off a human dispatcher cannot reliably make at scale, especially once unloading time, cold-chain vehicle availability, and a mixed fleet's different per-km costs are all in play simultaneously.

Phase 1 scope, explicitly

For this phase, pr_i is a static, category-based constant (not yet produced by Layer 2). The MILP is being built and validated against this constant so the optimization core, constraints, and solver pipeline are proven *before* wiring in the dynamic priority score – avoiding a hard dependency that would block progress.

4 Innovation & Technical Novelty

- **Context-aware demand forecasting, not generic time-series forecasting.** Encoding exam periods and heatwave/weather alerts as first-class model features is unusual even in commercial forecasting tools, and directly tailored to observable Tunisian consumption patterns (stimulant/analgesic demand during exams, rehydration/antihistamine demand during heat-waves).
- **Priority as an optimization coefficient, not a business rule.** Most routing software optimizes purely for cost and time. StockCare makes clinical/logistical urgency a first-class term inside the MILP objective, so the routing decision itself reflects medical need, not just geography – and the priority score's weights self-tune over time via reinforcement learning as real-world outcomes come in.
- **A genuinely closed loop.** Forecast → prioritize → route → deliver → update stock → forecast again. Each of the three layers is independently useful, but the compounding effect (accurate demand signal feeding a well-prioritized, well-routed fleet) is where the real gain sits.
- **Modular by design.** Layer 2's priority score is a pluggable interface ($pr_i \in [0, 1]$); Layer 3 already runs on a constant placeholder, so the team can ship, test, and improve each layer independently and swap in better models over time without re-architecting the system.

5 Originality

Individually, demand forecasting, medicine criticality scoring, and vehicle routing all exist elsewhere as separate tools. What does not currently exist – in the Tunisian pharmaceutical distribution chain, or in the generic logistics tools available on the market – is the **combination**, purpose-built for this exact context:

- Generic pharmacy ERPs/POS systems used by Tunisian officines track stock and sales; they do not forecast contextual demand or talk to a depot's routing system.
- Generic route-optimization software (fleet/logistics SaaS) optimizes cost and time; it has no notion of medicine criticality or cold-chain urgency as a routing input.
- PCT/depot allocation today is manual and largely reactive to what is asked for, not to a data-

driven, system-wide priority ranking across all pharmacies at once.

StockCare's originality is in treating pharmacy, depot, and delivery fleet as one connected optimization problem instead of three separate manual processes – and in doing so specifically for Tunisia's distribution structure and its documented stress points (PCT debt, currency-driven import constraints, chronic shortages) [3,4,5].

6 Quantified Impact

These are **projected targets for a pilot**, grounded in benchmarks from comparable AI-forecasting and route-optimization deployments elsewhere – not a guarantee, and not (yet) measured on Tunisian data. Phase 2 validates them against a real pilot depot and its pharmacies.

KPI	Projected target	Basis
Forecast error (vs. current manual reordering)	–20% to –50%	Published range for AI-driven demand forecasting vs. traditional methods [6]
Stockout incidents at pilot pharmacies	up to –65%	Industry benchmark for AI forecasting impact on stockouts [6]
Excess / expired inventory	–20% to –30%	Benchmark range for AI-guided inventory optimization [6]
Delivery distance & fuel use for depot fleet	–15% to –25%	Typical range reported for VRP/route-optimization deployments [7]
CO ₂ emissions from delivery fleet	–15% to –25%	Directly proportional to fuel-use reduction above [7]
Delivery latency for high-priority / cold-chain orders	internal target –30% vs. unranked manual dispatch	Project target – to be measured in pilot (no external benchmark; priority-weighted routing is the novel element)

Table 2: Projected Phase-1 impact targets, to be validated with real pilot data in Phase 2.

Honesty check

None of the rows above are Tunisia-specific measurements yet – they are literature-grounded targets that justify why the approach is worth piloting. Phase 2 should include instrumentation to measure every one of these KPIs against a live baseline before/after rollout, so Phase-1 promises become Phase-2 evidence.

7 Feasibility

7.1 Data sources required

- **Pharmacy stock & sales** – via API/export from each pharmacy's information system (most Tunisian officines already run some digital point-of-sale/stock software).
- **Weather & heatwave alerts** – Institut National de la Météorologie (INM) public data/API.
- **Academic calendar** – university/exam-period calendars (publicly published each year).

- **National medicine reference data** – a catalogue of medicines sold in Tunisia, therapeutic class, and essential-medicine status, cross-referenced against Ministry of Health / PCT nomenclature. The DPM's public AMM registry (*AMMs par nom de spécialité*) is a directly usable starting point, holding 6,058 active AMMs at extraction time [8].
- **Fleet & geography** – depot vehicle capacities, refrigeration availability, and pharmacy coordinates for the routing engine.

7.2 Technical feasibility

Every component uses well-established, tractable techniques: gradient-boosted trees for tabular demand forecasting, a multi-criteria weighted scoring model for priority, and MILP solved with an off-the-shelf solver for routing. None of the three layers requires research-grade novelty to implement a working version – the novelty is in how they are wired together.

7.3 Technology stack

The platform is built as a set of containerized services, each in the language best suited to its job, communicating over REST APIs:

Component	Technology	Rationale
Backend / core services	Java (Spring Boot)	Robust, industry-standard stack for the pharmacy–depot business logic, order management, and API layer
Frontend (pharmacy & depot dashboards)	Angular	Mature TypeScript framework for the two web dashboards (pharmacy view and depot dispatch view)
MILP routing engine	Python + PuLP	Clean algebraic modeling of the VRP formulation; solver-agnostic (CBC by default, swappable to a commercial solver for larger instances)
Demand forecasting & priority scoring	Python (pandas, scikit-learn / LightGBM)	Standard ML tooling for tabular forecasting and the multi-criteria priority model
Data storage	Relational DB (e.g. PostgreSQL)	Stock, orders, medicine catalogue, and delivery history
Deployment & orchestration	Docker (docker-compose)	Each service ships as a container – reproducible dev environments, straightforward pilot deployment on a single host, and a clean path to scale-out

Table 3: Planned technology stack: Java backend, Angular frontend, Python (PuLP) optimization core, Dockerized services.

This split keeps a deliberate seam between the *transactional* world (Java services handling stock, orders, and notifications) and the *analytical* world (Python services handling forecasting, priority scoring, and route optimization) – the two sides evolve independently and talk through well-defined APIs, which is exactly what the modular, swappable design in Section 3 requires.

7.4 Phased rollout

1. **Pilot scope:** one depot and a handful of partner pharmacies in a single region, a limited medicine subset (e.g. top-selling chronic + high-shortage-risk medicines).
2. **Static-priority MILP first** (current phase): prove the routing core works end-to-end with a constant priority weight.
3. **Dynamic priority integration:** swap in the priority classification model once ready, re-validate routing quality.
4. **Scale-out:** extend region coverage, medicine catalogue, and fleet size.

8 Target Audience

- **Primary:** private pharmacies (officines) and hospital pharmacies across Tunisia, who gain automatic restock alerts instead of reactive, manual reordering.
- **Primary:** regional depots and wholesale distributors (including PCT-adjacent structures), who gain a prioritized, optimized view of what to procure and how to deliver it.
- **Secondary / oversight:** the Ministry of Health and medicine regulators, who benefit from system-wide visibility into where shortages are emerging in near real time.
- **Ultimate beneficiary:** patients, particularly those on chronic treatment, who are the ones actually harmed by a stockout today.

9 Competitive Differentiation

Capability	Generic pharmacy ERP/POS	Generic logistics/VRP software	StockCare
Context-aware demand forecasting (exam periods, heatwaves, etc.)	No	No	Yes
Medicine-specific priority/criticality scoring	No	No	Yes
Priority as a routing optimization input	No	Cost/time only	Yes
Automatic pharmacy → depot restock query	No	N/A	Yes
Closed loop (forecast → prioritize → route → feedback)	No	No	Yes
Tailored to Tunisia's PCT/depot structure	N/A	No	Yes

Table 4: No single existing tool covers all three layers together for this specific market.

10 Business Model

The model is built around a simple principle: pharmacies and depots pay less for StockCare than they currently lose without it, so the subscription is framed as protecting an existing loss rather than as a new cost.

Value delivered. A pharmacy loses roughly 107 TND/month in margin per chronic patient it fails to retain (a 3-month refill of ~800 TND at a 40% margin). For a mid-size depot fleet, the platform's routing (a ~20% distance reduction) plus reduced spoilage yields on the order of 1,250 TND/month in fuel and cold-chain savings – visible in the first fuel bill, with no new trucks, stock, or staff.

Segment	Price	Trial	Value protected / saved
Pharmacy	100 TND/mo	1 month free, install included	~960 TND/mo margin at risk from stockouts
Depot	1,100 TND/mo	3 months free, install + training	~1,250 TND/mo fuel + spoilage savings
Transaction	0.5% of order value	—	Applied to orders routed through the platform

Table 5: Pricing anchored below the loss each segment already absorbs; free trials remove adoption risk.

Unit economics & moat. With an assumed 30% trial-to-paid conversion, LTV : CAC is favorable on both sides (roughly 16 : 1 for pharmacies, 20 : 1 for depots) with payback under ~2 months. The defensible asset is the proprietary, compounding dataset: the more pharmacies and depots the system connects, the better its forecasts and priority scores become, and the harder it is to replicate. Rollout follows the fleet's geography – starting in Sousse (Year 1), expanding across the Sahel, then Grand Tunis, Sfax, and the remaining governorates toward national coverage by Year 5.

1. Lock down the static-priority MILP against a synthetic or small real pharmacy/depot dataset.
2. Design and start training the priority classification model against the national medicine reference database.
3. Finalize the business model and pilot partner outreach.
4. Build the demand-forecasting model's feature pipeline (weather API, academic calendar, historical sales) and connect it to a first partner pharmacy's stock data.
5. Integrate all three layers end-to-end on the pilot scope; instrument the KPIs listed in Section 6 to replace projected targets with measured pilot results.

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